

1. Administrative & Study Identification

Full Study Title: A Phase Ib, Multi-arm, Open-label, Study of HDM201 in Combination With MBG453 or Venetoclax in Adult Subjects With Acute Myeloid Leukemia (AML) or High-risk Myelodysplastic Syndrome (MDS) **Short Title/Acronym:** HDM201 in Combination With MBG453 or Venetoclax in Patients With AML or High-risk MDS **Protocol Number:** CHDM201H12101C **NCT Number:** NCT03940352 **Posting ID:** 734407662148635645 **Sponsor Name:** Novartis (Novartis Pharmaceuticals) **Investigational Product:** HDM201 (siremadlin) in combination with MBG453 (sabatolimab) or Venetoclax (Venclexta) **Phase of Development:** PHASE1 (Phase Ib) **Medical Monitor:** Novartis Clinical Operations Lead **Preferred UMLS Name:** Myeloid Neoplasms

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, incidence and severity of Adverse Events (AEs) and Serious Adverse Events (SAEs), and the incidence of dose-limiting toxicities (DLTs), and to determine the Maximum Tolerated Dose (MTD) and/or Recommended Dose (RD) of HDM201 in combination with MBG453 or venetoclax in AML and high-risk MDS subjects. **Secondary Objectives:** To evaluate preliminary antitumor activity including Overall Response Rate (ORR), Best Overall Response (BOR), Event Free Survival (EFS) for AML, and Progression-Free Survival (PFS) for MDS. To characterize the Pharmacokinetics (PK) and Pharmacodynamics (PD) of the combinations.

2.2 Background & Rationale To address the unmet medical need in patients with relapsed/refractory AML or high-risk MDS who are ineligible for intensive chemotherapy. Siremadlin (HDM201) is an orally bioavailable, selective inhibitor of the p53-MDM2 interaction. Activation of the tumor suppressor protein p53 induces cell-cycle arrest and apoptosis in wild-type TP53 (TP53wt) tumors. This study aims to investigate if combinations with a TIM-3 targeting monoclonal antibody (MBG453/sabatolimab) or a Bcl-2 inhibitor (venetoclax) can broaden preliminary antileukemic efficacy, combat T-cell exhaustion, and overcome apoptotic evasion mechanisms.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Multi-arm, parallel group, non-randomized) **Blinding:** Open-label **Randomization:** N/A (Parallel assignment to treatment arms)

3.2 Planned Enrollment & Duration Targeted Enrollment: 52 subjects
Number of Sites: Multicenter (Global) **Estimated Study Start:** 24-Jun-2019 **Trial Status:** TERMINATED (Study was terminated early)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients ≥ 18 years of age at the date of ICF signature. 2. Present with one of the following: a. Relapsed/refractory AML following ≥ 1 (but ≤ 3) prior therapies who have relapsed or exhibited refractory disease (primary failure) and are deemed not to be candidates for standard therapy, including re-induction cytarabine/chemotherapy or hematopoietic stem cell transplantation (HSCT). b. First line AML patient unfit for standard induction chemotherapy (except where approved therapies are available); suitable/willing candidates for HSCT are excluded. c. High-risk MDS patient (high and very high-risk groups according to rIPSS) who have failed hypomethylating agent therapy. 3. ECOG performance status ≤ 1 . 4. TP53wt tumor characterized by, at minimum, no mutations in exons 5, 6, 7, and 8 of the TP53 gene, obtained from a bone-marrow sample collected no longer than 3 months before signing the main ICF. 5. Patient must be a candidate for serial bone marrow aspirate and/or biopsy according to institutional guidelines.

4.2 Exclusion Criteria 1. Prior combination treatment with compounds having the same mode of action (mdm2 or mdm4 inhibitors combined with TIM-3 inhibitors for arm 1, or Bcl-2 inhibitors for arm 2). 2. History of severe hypersensitivity reactions to any ingredient of study drug(s) and other monoclonal antibodies (mAbs) and/or their excipients. 3. Patients with acute promyelocytic leukemia with PML-RARA. 4. Allogeneic stem cell transplant (HSCT) within the last 6 months and/or active GvHD requiring systemic immunosuppressive therapy. 5. GI disorders impacting absorption of oral HDM201 or venetoclax. 6. Evidence of active bleeding or bleeding diathesis or major coagulopathy (including familial). 7. Patients with active, known, or suspected autoimmune disease (treatment arm 1 only).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Treatment Arm 1 (HDM201 + MBG453):** HDM201 oral capsules once daily on the first 5 days of a 28-day cycle; MBG453 IV infusion on Days 1 and 15 of the 28-day cycle. * **Treatment Arm 2 (HDM201 + Venetoclax):** HDM201 oral capsules once daily on the first 5 days of a 28-day cycle; Venetoclax (Venclexta) tablets oral daily (ramp-up dose over 4 to 5 days to achieve a 400 mg daily target dose) during the 28-day cycle.

Route of Administration: Oral (HDM201, Venetoclax) and Intravenous (MBG453). **Duration of Treatment:** Repeated 28-day treatment cycles until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for AML/MDS. **Prohibited:** Concurrent investigational agents, standard re-induction chemotherapy, or hematopoietic stem cell transplantation.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -90 to -1	Informed Consent, Medical History, TP53wt Bone Marrow sequencing, baseline labs
Baseline	Cycle 1 Day 1	First Dose, PK/PD blood sampling, Safety baseline
Interim	Days 1-5, 15 (C1-C+)	Drug administration (IV infusions on D1, D15 for Arm 1; ramp-up D1-5 for Arm 2), DLT monitoring, Safety Labs, Efficacy Assessment, Bone Marrow biopsy
End of Study	Follow-up	Final Safety checks, Post-treatment survival follow-up (up to Month 24)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Incidence of Adverse Events (AEs) and Serious Adverse Events (SAEs)** as a measure of safety (assessed up to Month 24 as study end). * **Severity of Adverse Events (AEs) and Serious Adverse Events (SAEs)** as a measure of safety (assessed up to Month 24 as study end). * **Incidence of dose-limiting toxicities (DLTs)** of treatment by the end of the first cycle, to determine the MTD and/or Recommended Dose (RD) of HDM201 in combination with MBG453 or venetoclax.

7.2 Secondary Outcomes * Preliminary antitumor activity: Overall Response Rate (ORR) and Best Overall Response (BOR). * Event Free Survival (EFS) for AML (Cheson 2003) or Progression Free Survival (PFS) for MDS (Cheson 2006). * Pharmacokinetics (PK) parameters (e.g., Cmax, AUC) and Pharmacodynamics (PD).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during the 28-day cycles up to Month 24. Thrombocytopenia is actively tracked as an on-target effect of MDM2 inhibition. **Serious Adverse Events (SAEs):** Routine SAE reporting timelines apply. DLT observation window operates strictly through the first 28-day cycle. **Data Monitoring Committee (DMC):** Internal Dose Escalation Committee responsible for reviewing safety data between cohorts prior to approving escalation or defining the MTD.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving at least one dose of study drug) and Efficacy/PK Analysis Set. **Sample Size Justification:** A formal power calculation is not applicable for this Phase Ib dose-escalation design; patient cohorts expanded incrementally based on Bayesian or standard dose-escalation algorithms until MTD was reached (targeted enrollment of 52). **Handling of Missing Data:** Missing data handled as per standard Phase I protocol (mostly descriptive analysis of observed data with minimal imputation).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical site monitoring, centralized data review, and dose-escalation committee oversight. **Audit Trail:** Assured via standard eCRF systems, query generation, and central laboratory verification of TP53 wild-type sequencing.

11. Study Identifiers & Governance

Posting ID: 734407662148635645 **Primary ID:** CHDM201H12101C
Registry Number: NCT03940352 **Sponsor/Lead Organization:** Novartis
Study Title: HDM201 in Combination With MBG453 or Venetoclax in Patients With Acute Myeloid Leukemia (AML) or High-risk Myelodysplastic Syndrome (MDS) **Official Regulatory Title:** A Phase Ib, Multi-arm, Open-label, Study of HDM201 in Combination With MBG453 or Venetoclax in Adult Subjects With Acute Myeloid Leukemia (AML) or High-risk Myelodysplastic Syndrome (MDS)
Status: TERMINATED (Study was terminated early)

12. Clinical Framework (AML/MDS Specific)

Primary Conditions: Acute Myeloid Leukemia (AML), High-risk Myelodysplastic Syndrome (MDS), Myeloid Neoplasms. **Diagnosis Threshold:** Relapsed/refractory disease post 1-3 therapies, unfit first-line AML, or high-risk MDS failed on HMAs, accompanied by confirmed TP53wt mutation-free exons 5-8. **Medical Methodology:** Targeted dual therapy (MDM2-p53 inhibition + TIM-3 blockade OR Bcl-2 inhibition) **Optimization Target:** Identifying the highest tolerable combined dose (MTD) mitigating leukemic resistance.

13. Study Procedures & Methodology

Management Pathway: Serial bone marrow aspirate and/or biopsy mandatory at screening, during, and at the end of therapy

according to institutional guidelines. **Education Protocol (HCP):** DLT identification, Venetoclax ramp-up monitoring. **Education Protocol (Patient):** Oral medication adherence instructions for HDM201 and Venetoclax dosing. **Quality Audit Mechanism:** Centralized genomic screening for TP53 wild-type status before enrollment.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or treatment intolerance, followed by follow-up extending up to Month 24.
Onsite Visit Cadence: Frequent during Cycle 1 (Days 1-5 for ramp-up/dosing; Days 1, 15 for IV infusions in Arm 1). **Remote Monitoring Frequency:** Standard safety check-ins as defined by local protocols. **Checklist Review Interval:** Conducted each 28-day cycle for AE reporting and dose modification assessments.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				